



Impact of energy sector volatility on clean energy assets

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ABSTRACT

We study the effect of uncertainty in energy sector firms, as reflected in the information in the energy sector implied volatility index (VXXLE), on clean energy exchange traded funds (ETFs). Unlike the existing literature that mainly employs linear models to study the association between clean energy and financial markets, we apply the Markov regime switching process to uncover the impact of the VXXLE on clean energy ETFs during high and low volatility regimes. Our findings reveal a negative impact, suggesting that, when implied volatility levels are high for energy sector firms, a drop is likely in clean energy asset returns. The results also show evidence of an asymmetric effect. During high volatility regimes, the association between the VXXLE and clean energy ETFs holds stronger than in low volatility regimes. These results remain unchanged controlling for the effects of oil prices, crude oil volatility, and technology stock prices. We further document that changes in levels of the VXXLE substantially impact the realized volatility of clean energy ETFs. Our findings have significant implications for investors and policymakers.

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1. Introduction

In recent years, investment in the global clean energy industry has seen an upward trend. For example, investments in this industry amounted to \$332.1 billion in 2018. In fact, investment in renewable energies has been over \$300 billion for the last five years in a row. During the period 2010–2018, \$2.6 trillion has been spent in the world alternative energy sector, more than treble the amount spent in the previous decade (2000–2010).¹

Considering the significance of the clean energy market as a new asset class, the academic literature on the global clean energy sector is growing but remains embryonic in several aspects. Firstly, most existing studies consider oil price returns and clean energy

stock indices (e.g., [8,23,33,10,34,2,11,41,38,36]). Secondly, few studies consider the impact of energy-related volatility indices such as the crude oil implied volatility index (OVX), on clean energy stocks (e.g., [12,1,15]). However, the information in the energy sector implied volatility index (henceforth, VXXLE), published by the Chicago Board Options Exchange (CBOE), reflects energy sector risk. Although the VXXLE can be viewed as a superior measure for tracking the risk associated with energy sector companies, it does not receive much attention in academic studies dealing with clean energy stocks. To the best of our knowledge, only a few studies examine the behaviour of this implied volatility index. López [1] shows that the VXXLE experiences a drop in response to US macroeconomic news and events related to OPEC meetings. Nikkinen and Rothovius [2] argue that the VXXLE can be decomposed into two components related to OVX and VIX, with 55% and 45% weights, respectively.

Against this background, the aim of this paper is to assess the impact of the VXXLE on clean energy exchange-traded funds (ETFs). To serve our purpose, we consider several energy ETFs and apply regime switching models, accounting for various control variables.

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¹ Specifically, China has invested \$758 billion over the past 10 years followed by the US (\$356 billion) and Japan (\$202 billion). During this period, Europe as a whole invested \$698 billion (see Bloomberg NEF).

We also examine the risk transmission between the VXXLE and clean energy ETFs.

Our contributions are as follows. Firstly, we are the first to assess the effect of the VXXLE on the stock market performance of clean energy firms. Given that clean energy assets can be highly volatile [3], it is important for economic actors to understand the dynamics and sources of such risks so that appropriate decisions regarding portfolio construction and risk management can be taken. Notably, the use of the VXXLE could be informational, as this volatility index characterizes a forward-looking measure of uncertainty in the stock prices of energy sector companies following the VIX methodology [1]. In fact, Wen et al. [4] argue that investors holding assets in clean energy equity markets closely follow the equity prices of fossil fuel companies given that both equities are generally viewed as competing assets. Wen et al. [4] also document that the equity prices of renewable energy firms are driven by both positive and negative news stemming from the energy sector equity market. In particular, investors may worry that once positive news about fossil fuel equities emerges, investment funds largely switch from new energy equities to the fossil fuel equities, thus dampening their returns. As the VXXLE reflects the 30-day market uncertainty about energy companies' returns, one can postulate that the information in the VXXLE could have substantial predictive power for the equity prices of clean energy companies. Hence, our analyses have important implications for market participants who are keen to predict energy firms' future uncertainty.

Secondly, in contrast to the existing literature, we make use of clean energy ETFs instead of clean energy equities. Doing so is advantageous as these ETFs are extremely liquid and easily traded like stock [5]. In addition, ETFs are free from the non-synchronous trading issues related to stock index data. Lo and MacKinlay [6]; for example, argue that avoiding non-synchronous trading problems can lead to more robust results related to market efficiency tests.

Thirdly, we use the Markov regime switching (MRS) model to study the effect of the VXXLE on clean energy ETFs. Existing studies focusing on the association between renewable energy investment and other financial assets mostly employ linear models (e.g., ordinary least squares regression), albeit the linkage among financial markets is prone to recurrent changes as a consequence of economic shocks, terrorist attacks, and natural disasters.² Therefore, the fixed parameter assumption of the least squares method is too restrictive, and hence the applied model could be misspecified. To this end, adopting the MRS regression is beneficial given that its specification allows the estimated coefficients to swing across a distinct number of states [7]. By employing the MRS regression, we explore how clean energy assets react to energy market volatility shocks during high and low volatility states.

Fourthly, we examine whether energy sector uncertainty indices (VXXLE and OVX) impact the volatility of clean energy ETFs. This is important for investors and policymakers as it allows them to understand the risk transmission relationship between these markets, which is important for option pricing and risk spillover inferences.

The next section reviews the related literature. Section 3 describes the dataset. The section that follows outlines the econometric models. Results are presented and analysed in Section 5, while Section 6 discusses the findings. Section 7 concludes.

1.1. Related studies

A main strand of literature considers clean energy stock indices and their relationship to other assets, especially crude oil. Sadorsky [8] documents a positive association between oil returns and clean energy equity returns and evidence of significant volatility transmissions. Kumar et al. [9] use a panel data regression and show that clean energy equities are sensitive to energy price fluctuations. Reboredo [10] finds that the association between crude oil prices and clean energy prices tends to vary over time [11], provide evidence of a unidirectional linkage between oil and clean energy indices, mainly running from oil prices to equity prices. Employing wavelets, Reboredo et al. [12] conclude that the association between oil and alternative energy stocks holds stronger in the long-term. In addition, Ahmad [13] concludes that oil and clean energy assets are positively correlated, suggesting that higher oil prices would lead to an upturn in the return of clean energy stocks. Ferrer et al. [14]; however, find that the correlations between oil and clean energy stock prices are lower than that between technology and clean energy stock indexes. Recently [15], report that important commodities such as crude oil and precious metals act as safe-haven assets for alternative energy markets during bearish periods. Lin and Chen [16] also document that energy asset can be used to hedge the risk of new energy stocks in China. Xia et al. [17] find that fluctuations in electricity, oil and coal prices lead to a substantial change in clean energy equity returns. Yahya et al. [18] use the cross-quantile correlation approach and show that clean energy stock indices are influenced by fluctuations in nonferrous metal prices. Sun et al. [19] reveal that oil prices account for only a small part of stock price fluctuations of new energy companies.

Another strand of literature deals with risk transmission involving clean energy stocks. Dutta et al. [20] demonstrate that the risk of clean energy equities could be diversified using the information contents of the European Union emission trading prices. Dutta [21] shows that uncertainty in the silver market, which is used in the photovoltaic system for generating solar energy, has a significant negative impact on solar energy firms' equity prices. Maghyreh et al. [22] combine wavelets with the multivariate GARCH model to evidence a bidirectional volatility transmission relationship between oil and clean energy stock markets. Saeed et al. [23] use the DCC-GARCH process and show that clean energy stocks can hedge oil price risk. Dutta et al. [24] report that green assets are more susceptible to oil market volatility than oil price fluctuation. Liu and Hamori [25] document that the oil market transmits volatility to the US and European renewable energy stock markets.

Note that few studies use the information content of the OVX to assess whether crude oil volatility affects clean energy equity returns. Dutta [26]; for example, finds that the OVX and the realized volatility of clean energy equities move in tandem. Ahmad et al. [3] indicate that the OVX and clean energy equity indices are inversely related, suggesting the ability of the OVX to diversify the risk of clean energy stock markets.

In this paper, we extend the above lines of research by considering for the first time the impact of the VXXLE on clean energy ETFs. We apply regime switching models and account for various control variables. Few studies employ regime switching models to investigate the linkages between clean energy or green assets and other financial markets. Managi and Okimoto [27] employ the regime switching process to detect the possibility of a structural change by analysing smoothed probabilities. The authors find regime shifting behaviour after 2007 which, in turn, impacts the linkages between oil and clean energy stock markets. Furthermore, the work of Dutta et al. [24] adopts the MRS model and indicates a

² Only a couple of studies [24,27] employ regime switching models to investigate the linkage between clean energy or green assets and other financial markets [27]. employ the regime switching process to detect the possibility of a structural change by analyzing smoothed probabilities. The authors find regime shifting behavior after 2007 which, in turn, impacts the linkage between oil and clean energy stock markets. The work of [24] adopts the MRS model and indicates a swapping between regimes implying that there exist high and low volatility states for green assets.

swapping between regimes, implying that there exist high and low volatility states for green assets. Our empirical analysis also involves the risk transmission between the VXXLE and clean energy ETFs.

2. Data

Three clean energy ETFs are considered in this study: Invesco WilderHill Clean Energy ETF (henceforth, PBW), Invesco Global Clean Energy ETF (henceforth, PBD) and iShares Global Clean Energy ETF (henceforth, ICLN). PBW allows investors exposure to clean energy investment. PBD offers a global perspective on exposure to the clean energy index. ICLN offers a way to invest in companies comprising various sub-sectors related to wind and solar energy sources. In addition, we use the VXXLE index to reflect the risk of energy sector firms. Computed by the CBOE, the VXXLE reflects the 30-day market uncertainty on energy companies' returns, using the VIX methodology for options on the Energy Sector Select SPDR Fund (XLE), spanning a wide range of strike prices. Daily data from the VXXLE and various clean energy ETFs are extracted from DataStream. They cover the period March 16, 2011–December 31, 2018, yielding 2034 daily observations. Fig. 1 shows the time-series evolution of all indices under study, in which a significant drop is observed in energy sector asset prices for the period of 2014–2016, when the crude oil market experienced a downturn due to the oversupply of crude oil.

The empirical analyses are conducted with daily log returns. Table 1 shows that all clean energy assets experience negative returns during the full sample period. The VXXLE shows a higher volatility than the clean energy indices. All return series exhibit nonzero skewness with kurtosis exceeding 3, suggesting non-normality in the return distributions, which is validated using the Jarque-Bera test.

We employ the augmented Dickey-Fuller (ADF) and Phillips-Pearson (PP) tests to examine whether the data used are stationary. The results, shown in Table 2a, confirm the rejection of the null hypothesis of non-stationarity for all the return series. Given the non-normality of the return distributions, we apply the RALS-LM unit root test, and present the findings in Table 2b, confirming that the return indices are stationary.

Table 3 shows the unconditional correlations among the return series. The correlations among the clean energy ETFs appear to be positive, whereas each index is negatively related to the VXXLE. Such an inverse relationship implies that portfolio risk can be reduced by holding assets in these markets.

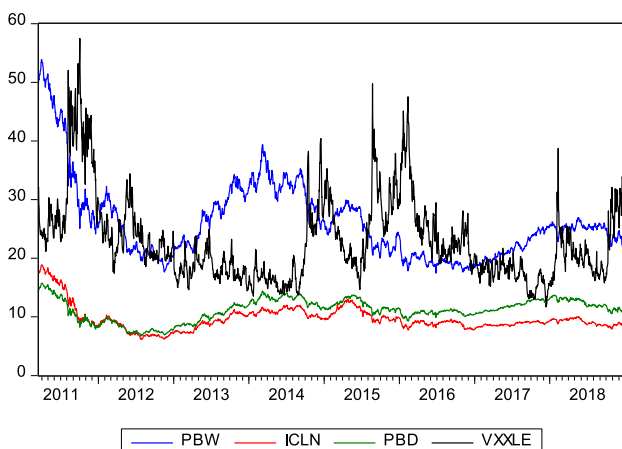


Fig. 1. Time-series plot of the indices studied.

Table 1
Descriptive statistics.

Index ↓	Mean	Std. Dev.	Skewness	Kurtosis	Jarque-Bera test
PBW	−0.0182	0.715	−0.277	5.567	584.07***
PBD	−0.0069	0.582	−0.377	7.101	1472.69***
ICLN	−0.0158	0.637	−0.336	5.901	751.08***
VXXLE	0.0004	1.511	0.951	12.371	7745.54***

Notes: This table reports the descriptive statistics of the indices under study. For clean energy ETFs we consider the logarithmic differences, while for the VXXLE we use simple differences. *** indicates statistical significance at 1% level.

Table 2a
Results of unit root tests.

Index ↓	ADF Tests		PP Tests	
	Level	1st Difference	Level	1st Difference
PBW	−3.75*** (.00)	−26.67*** (.00)	−3.52*** (.00)	−42.87*** (.00)
PBD	−2.12 (.24)	−29.89*** (.00)	−2.14 (.62)	−45.38*** (.00)
ICLN	−3.88*** (.00)	−26.53*** (.00)	−3.59*** (.61)	−45.17*** (.00)
VXXLE	−4.74*** (.00)	−24.17*** (.00)	−4.22*** (.75)	−49.60*** (.00)

Notes: p-values are given in parentheses. *** indicates statistical significance at 1% levels.

Table 2b
Results of RALS-LM unit root tests.

	Test statistic	$\hat{\rho}^2$	$\hat{T}_B$	$\hat{k}$
PBW	12.26***	0.259	2011	2
PBD	5.89***	0.914	2011	5
ICLN	6.02***	0.787	2011	1
VXXLE	6.88***	0.558	2011	0

Table 3
Correlation matrix.

	PBW	PBD	ICLN	VXXLE
PBW	1			
PBD	0.8130***	1		
ICLN	0.7677***	0.8198***	1	
VXXLE	−0.6162***	−0.6383***	−0.5989***	1

Notes: This Table shows the unconditional correlations among the indices. *** indicates statistical significance at 1% levels.

3. Methodology

3.1. Markov regime switching (MRS) approach

The MRS model receives enormous attention in the finance literature (e.g., [7,6,39,9]. Uddin et al. [7] argue that financial time-series are nonlinear in nature and therefore it is important to use econometric models which take into account the asymmetric association among the financial time-series. To this end, employing the MRS model is beneficial, as it captures such asymmetry and allows the parameters to shift between various regimes. Accordingly, our MRS model has the following form:

$$R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t} R_{i,t-1} + \gamma_{i,r_t} \Delta \left(\frac{\pi}{2} - \theta \right) VXXLE_t + u_{i,t} \quad (1)$$

where $R_{i,t}$ is the log return for i -th clean energy ETF index at time t , r_t refers to a discrete regime variable, α_{i,r_t} is the regime-dependent intercept, and β_{i,r_t} and γ_{i,r_t} are regime-dependent slope coefficients. At the current period t , the diffusion probability from regime 1 to regime m at the next period $t + 1$ is entirely dependent on the regime at the current period t :

Table 4
Results of Markov regime switching model.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	VXXLE shock	Sigma	χ^2 test	
PBW	S1	−0.031 (0.28)	0.025 (0.40)	−0.301*** (0.00)	0.733*** (0.00)	177.51***	
	S2	−0.009 (0.51)	0.073** (0.03)	−0.276*** (0.00)	0.445*** (0.00)		
PBD	S1	0.010 (0.27)	0.034 (0.11)	−0.257*** (0.00)	0.755*** (0.00)	178.02***	
	S2	−0.114** (0.02)	−0.108** (0.03)	−0.234*** (0.00)	0.357*** (0.00)		
ICLN	S1	0.042** (0.03)	0.016 (0.44)	−0.319*** (0.00)	0.577*** (0.00)	116.39***	
	S2	0.015 (0.35)	−0.059* (0.09)	−0.150*** (0.00)	0.375*** (0.00)		
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9835	0.0165	0.0093	0.9907	60.71	107.91	21.82
PBD	0.9811	0.0189	0.0977	0.9023	52.91	10.24	21.93
ICLN	0.9833	0.0167	0.0118	0.9812	60.15	53.14	22.41

Notes: The results are based on the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t} R_{i,t-1} + \gamma_{i,r_t} \Delta VXXLE_t + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

$$p_{jk} = \Pr(r_{t+1} = k | r_t = j), p_{jk} \geq 0, \sum_{k=1}^M p_{jk} = 1 \quad (2)$$

To evaluate the regime-dependent parameters, Equation (1) is estimated for two regimes. Our objective is to examine whether the association between energy sector volatilities and clean energy ETFs holds stronger in the high volatility regime than the low volatility regime. Moreover, following previous studies [3,39], we consider estimating the regime classification measure (RCM) to gauge the accuracy of the MRS approach:

$$RCM(r) = 100r^2(1/T) \sum_{t=1}^T \prod_{i=1}^r \hat{p}_{i,t} \quad (3)$$

The RCM statistics range from 0 to 100, where 0 implies that the MRS model classifies the regimes precisely, while 100 indicates that the model fails to do so.

3.2. Test of asymmetric effects

We further examine the existence of asymmetric relations between the VXXLE and clean energy ETFs, which is crucial as asset prices often behave asymmetrically to changes in market uncertainty [24]. Information on the presence of a symmetric association would allow clean energy firms to properly gauge the risk emanating from the energy sector, while managing such risk could be difficult if the impact of the VXXLE is nonlinear. Because increases and decreases in the VXXLE might cause cyclical fluctuations in clean energy business, exploring the asymmetric impact of such shocks on clean energy assets is of paramount importance.

For the inspection of the asymmetric relationship, the following mean equation is estimated:

$$R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t} R_{i,t-1} + \gamma_{1i,r_t} \Delta VXXLE_t^+ + \gamma_{2i,r_t} \Delta VXXLE_t^- + u_{i,t} \quad (4)$$

where $\Delta VXXLE_t^+ = \max(\Delta VXXLE_t, 0)$ and $\Delta VXXLE_t^- = \min(\Delta VXXLE_t, 0)$ refer to positive and negative energy volatility shocks. To examine the presence of asymmetric associations, it suffices to test $H_0 : \gamma_1 = \gamma_2$.

3.3. Risk transmission relationship

So far, we have considered whether energy sector uncertainty indices (VXXLE and OVX) impact the first-order moment (i.e., mean return) of clean energy ETFs. It is, however, important for investors and policymakers to understand the risk transmission from those energy sector uncertainty indices to the second-order moment (i.e., volatility) of clean energy ETFs, given that what matters for both market participants and policymakers is not market price variations per se, but their unpredictability and the resultant risks for producers, traders, consumers, and government agents.

In this section, we thus examine the effect of the VXXLE and OVX on the realized volatility (RV) of clean energy assets.³ For comparison purposes, the United States Oil Fund or USO ETF index is also considered. In addition, the effect of technology stock price volatility (RV_PSE) is measured. Note that the RV of each index is proxied by squared returns. To serve our purpose, we estimate the following models:

$$RV_t = \delta_0 + \delta_1 RV_{t-1} + \varepsilon_t \text{ (Baseline RV model)} \quad (5)$$

$$RV_t = \delta_0 + \delta_1 RV_{t-1} + \delta_2 \Delta VXXLE_t + \varepsilon_t \text{ (RV - VXXLE model)} \quad (6)$$

$$RV_t = \delta_0 + \delta_1 RV_{t-1} + \delta_2 \Delta OVX_t + \varepsilon_t \text{ (RV - OVX model)} \quad (7)$$

$$RV_t = \delta_0 + \delta_1 RV_{t-1} + \delta_2 RV_PSE_t + \varepsilon_t \text{ (RV - PSE model)} \quad (8)$$

where, RV_t denotes the realized volatility of the PBW/PBD/ICLN/USO index at time t .

4. Results

4.1. Results of MRS regression

Panel A of Table 4 presents the estimates of the MRS regression, whereas Panel B of Table 4 shows the transition probabilities and expected durations. A number of notable findings emerge. First, the VXXLE has a negative impact on the clean energy ETFs under study.

³ We also consider another measure of volatility based on the conditional volatility of GARCH models and the results remain qualitatively the same.

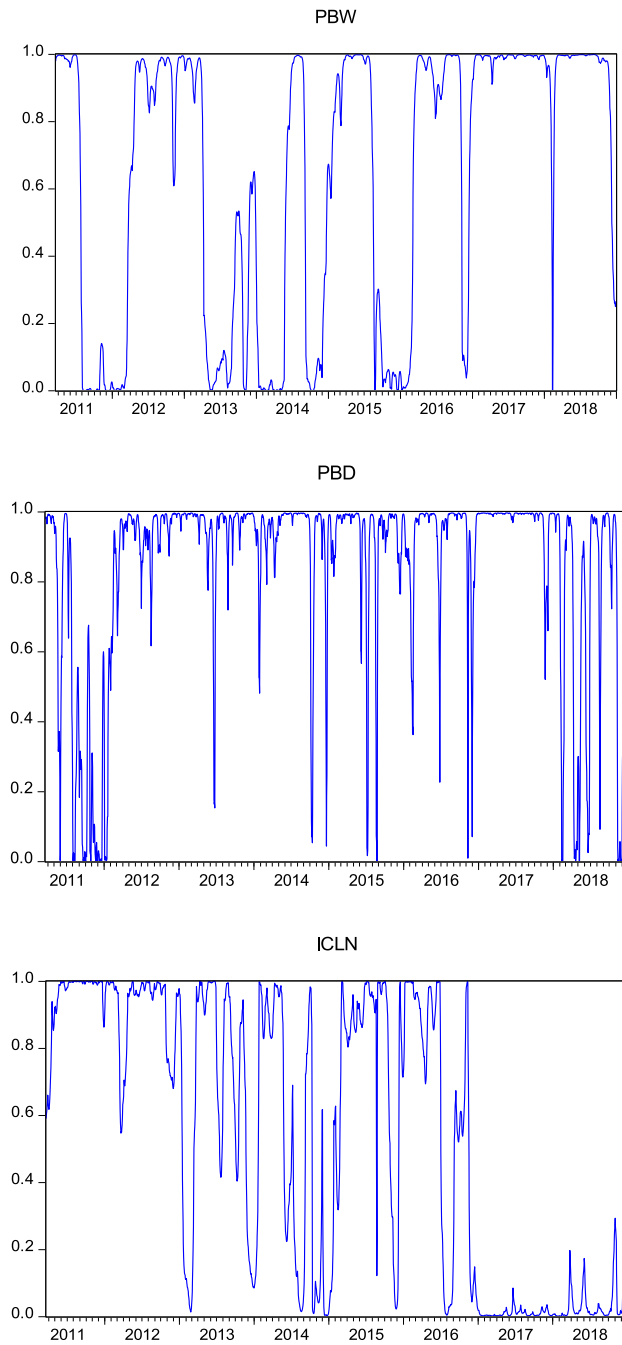


Fig. 2. Smoothed probabilities of being in high volatility states for various clean energy ETFs.

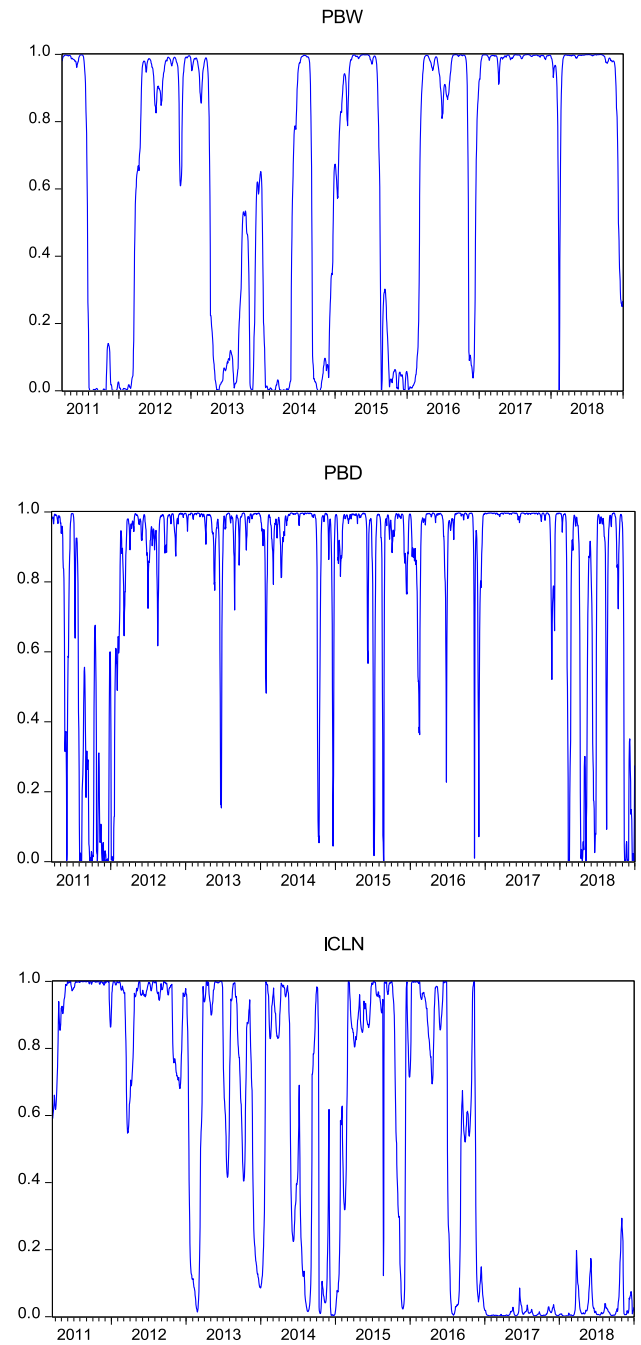


Fig. 3. Smoothed probabilities of being in low volatility states for various clean energy ETFs.

Thus, when volatility levels are high for energy sector companies, a drop is likely to be observed in clean energy ETFs. Such a finding is expected, given that an upsurge in energy sector volatility leads to a fall in crude oil prices, which, in turn, leads to a decline in the prices of clean energy ETFs [26]. Another possible reason could be that an inverse association usually exists between implied volatilities and stock returns. In fact, a strand of literature confirms this result. Badshah [28]; for example, documents a negative association between stock returns and changes in the implied volatility index for the US, Germany, and the UK markets. Additionally, Basher and Sadorsky [29] show that the VIX index is negatively related to emerging market stock indices. In line with previous literature, we

show that risk and return are inversely related for energy sectors.

Second, the sigma coefficient, which represents the magnitude of volatility measured by standard deviation for each regime, is higher for regime 1 than regime 2. This finding implies that regime 1 (regime 2) is the high (low) volatility regime. Moreover, all the sigma coefficients are statistically significant, implying a switch between high and low volatility regimes. We further note that the effect of the VXXLE is higher in regime 1 than regime 2. Therefore, for the high volatility regime the association between energy sector volatilities and clean energy ETFs holds stronger than for the low volatility regime.

Third, the DU1 and DU2 statistics representing the expected

durations for occupying a specific regime, suggest that the PBW index, in comparison to other series, has higher probabilities of being in a particular state. Additionally, based on the smoothed probability for all clean energy ETFs, both regimes seem to be highly persistent as the values of p_{11} and p_{22} are close to 1 in most cases.

Fourth, the RCM values indicate that the MRS regression fits well for each of the clean energy ETFs under study. Figs. 2 and 3 depict smooth probabilities belonging in high and low volatility regimes. Consistent with the RCM statistics reported in Table 4, a strong switching pattern between the regimes is observed for all clean energy assets. In particular, we notice that at the beginning of our sample period (i.e., March 2011) the probability of being in the high volatility state is very high (close to 1) for all the ETFs (see Fig. 2). Furthermore, Fig. 3 demonstrates that during the same period the probability of being in the low volatility state remains very low. One would expect that the global energy market becomes highly uncertain in March 2011, most probably due to the Libyan war, which, in turn, leads to lower stock prices. Moreover, there is a high probability of being in the high volatility regime throughout the year 2015 (see Fig. 2), when the energy market experiences a downturn due to the oversupply of crude oil. This finding is confirmed by Fig. 3, in which we observe a low probability of being in the low volatility regime.

Overall, the findings suggest that the energy sector implied volatility exerts a negative impact on clean energy assets. Our results are consistent with Dutta [21], who shows that oil and metal volatility indices have negative effects on solar energy stock returns. Ahmad et al. [3] also conclude that the OVX and clean energy equities are inversely related. One possible reason for this

inverse association between risk and return is that price decreases could raise the portfolio adjustment cost of risk-averse agents more than price increases [30].

4.2. Asymmetric effects

The findings of our asymmetric analysis, reported in Table 5, mimic those presented in Table 4. For example, the impact of the VXXLE on clean energy ETFs remains negative, which suggests that when energy sector volatility decreases, clean energy assets experience an increase in their prices. Alternatively, an upturn in energy sector volatility leads to a decrease in the prices of clean energy assets. It is noteworthy that, for the asymmetric analyses, regime 2 appears to be the high volatility state for all the indices under study. We also note that for all these indices, both negative and positive volatility shocks have significant effects on clean energy assets.

The results of the asymmetric tests ($H_0: \gamma_1 = \gamma_2$), are shown in Table 5, and imply that the effect of the VXXLE is asymmetric (with few exceptions). Hence, changes (upward and downward shifts) in energy sector volatility exert a diverse effect on the prices of clean energy ETFs.

The nonlinear associations between the sectors under study have dynamic implications for researchers and policymakers. For example, such associations encourage academics and risk managers to adopt nonlinear models instead of linear models. Furthermore, companies operating in new energy segments should also take a dynamic approach to hedging the risk of the clean energy sector. The asymmetric impact of energy sector volatility on clean energy assets should be considered in model-based specifications.

Table 5
Results of asymmetric effects.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	VXXLE Shock (+)	VXXLE Shock (−)	Sigma	$H_0: \gamma_1 = \gamma_2$
PBW	S1	0.023 (0.30)	0.068*** (0.00)	−0.309*** (.00)	0.232*** (0.00)	0.440*** (0.00)	−1.87*
	S2	−0.009 (0.82)	0.019 (0.52)	−0.317*** (.00)	0.281*** (0.00)	0.734*** (0.00)	−0.85
PBD	S1	0.043*** (0.00)	0.028 (0.20)	−0.258*** (.00)	0.190*** (0.00)	0.361*** (0.00)	−2.88***
	S2	−0.043 (0.56)	−0.126** (0.03)	−0.298*** (.00)	0.210*** (0.00)	0.771*** (0.00)	−1.14
ICLN	S1	−0.055 (0.25)	0.016 (0.73)	−0.213*** (.00)	0.206*** (0.00)	0.364*** (0.00)	2.45**
	S2	0.047** (0.02)	−0.026 (0.38)	−0.324*** (.00)	0.238*** (0.00)	0.741*** (0.00)	−3.77***
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9897	0.0103	0.0189	0.9811	97.45	52.97	21.87
PBD	0.9842	0.0158	0.0939	0.9061	63.45	10.85	22.01
ICLN	0.6312	0.3688	0.1599	0.8401	2.71	6.25	24.51
							χ^2 test
PBW							157.88***
PBD							145.82***
ICLN							222.86***

Table 6
Subsample analyses.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	VXXLE shock	Sigma	χ^2 test	
PBW	S1	−0.034 (0.41)	0.036 (0.49)	−0.234*** (0.00)	0.657*** (0.00)	53.52***	
	S2	−0.060* (0.06)	0.032 (0.56)	−0.427*** (0.00)	0.375*** (0.00)		
PBD	S1	0.003 (0.92)	−0.004 (0.95)	−0.099*** (0.00)	0.286*** (0.00)	5.74**	
	S2	−0.029 (0.29)	−0.005 (0.91)	−0.342*** (0.00)	0.386*** (0.00)		
ICLN	S1	−0.076 (0.28)	0.016 (0.89)	−0.168*** (0.00)	0.704*** (0.00)	35.75***	
	S2	0.018 (0.60)	−0.019 (0.77)	−0.311*** (0.00)	0.343*** (0.00)		
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9800	0.0200	0.0256	0.9744	50.11	39.12	25.02
PBD	0.6293	0.3703	0.2010	0.7990	2.69	4.97	24.89
ICLN	0.3235	0.6765	0.3763	0.6237	1.48	2.66	26.61

4.3. Additional analyses

4.3.1. Subsample analysis

This section scrutinizes the effect of crisis periods on the relationship between energy sector uncertainty and clean energy assets. To reach this goal, we conduct subsample analyses to examine

whether clean energy assets react differently to the VXXLE under diverse market circumstances. Specifically, we consider the period December 2014–March 2016, which covers the oil market downturn during which the oversupply of crude oil adversely affected crude oil prices.

The results of our subsample investigation are given in Table 6.

Table 7

Robustness test based on realized volatility.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	Realized volatility	Sigma	χ^2 test	
PBW	S1	0.110*** (0.00)	−0.068** (0.04)	−0.506*** (0.00)	0.677*** (0.00)	45.11***	
	S2	−0.100*** (0.00)	0.162*** (0.00)	−0.361*** (0.00)	0.432*** (0.00)		
PBD	S1	0.053*** (0.00)	−0.008 (0.79)	−0.488*** (0.00)	0.594*** (0.00)	106.46***	
	S2	−0.058** (0.03)	−0.018** (0.60)	−0.307*** (0.00)	0.319*** (0.00)		
ICLN	S1	−0.062** (0.03)	0.028 (0.44)	−0.275*** (0.00)	0.683*** (0.00)	164.11***	
	S2	0.043** (0.04)	−0.023 (0.43)	−0.511*** (0.00)	0.343*** (0.00)		
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.4694	0.5306	0.4229	0.5771	1.88	2.36	27.11
PBD	0.6888	0.3112	0.3102	0.6898	3.21	3.23	27.87
ICLN	0.7022	0.2978	0.3415	0.6585	3.35	2.93	28.64

Table 8

Robustness test based on GARCH volatility.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	GARCH volatility	Sigma	χ^2 test	
PBW	S1	0.019 (0.24)	0.043 (0.18)	−0.814** (0.04)	0.952*** (0.00)	296.82***	
	S2	−0.076** (0.04)	0.063* (0.08)	−0.645* (0.08)	0.492*** (0.00)		
PBD	S1	−0.096** (0.02)	−0.017 (0.69)	−0.798** (0.04)	0.895*** (0.00)	449.94***	
	S2	−0.033*** (0.00)	−0.002 (0.96)	−0.749** (0.03)	0.381*** (0.00)		
ICLN	S1	−0.119** (0.04)	0.033 (0.55)	0.744 (0.16)	0.942 (0.24)	257.79***	
	S2	0.006 (0.62)	−0.007 (0.81)	−0.646** (0.04)	0.486*** (0.00)		
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9820	0.0180	0.0271	0.9729	55.70	36.94	26.84
PBD	0.9486	0.0514	0.0213	0.9787	19.44	46.97	27.08
ICLN	0.9379	0.0621	0.0138	0.9862	16.10	72.35	27.96

Notes: This table presents the results of the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,t} + \beta_{i,t} R_{t-1} + \gamma_{i,t} CV_t + u_{i,t}$, where $R_{i,t}$ refers to the return for green assets at time t and CV_t is the conditional volatility of the XLE index at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * denote significance of results at 1%, 5% and 10% levels respectively.

Table 9

Impact of the VXXLE controlling for technology stock price shocks.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	PSE-100	VXXLE	Sigma	χ^2 test
PBW	S1	−0.065 (0.23)	0.032 (0.17)	0.968*** (0.00)	−0.300*** (0.00)	1.377*** (0.00)	162.95***
	S2	−0.027 (0.31)	0.061*** (0.00)	0.862*** (0.00)	−0.131*** (0.00)	0.852*** (0.00)	
PBD	S1	0.003 (0.86)	0.011 (0.43)	0.821*** (0.00)	−0.199*** (0.00)	0.683*** (0.00)	206.49***
	S2	−0.146 (0.15)	−0.070 (0.17)	0.463*** (0.00)	−0.240*** (0.00)	1.521*** (0.00)	
ICLN	S1	0.009 (0.73)	0.019 (0.47)	0.700*** (0.00)	−0.133*** (0.00)	0.491*** (0.00)	129.91***
	S2	−0.145** (0.04)	0.007 (0.43)	0.623*** (0.00)	−0.400*** (0.00)	0.587*** (0.00)	
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9853	0.0147	0.0086	0.9914	68.26	116.70	19.02
PBD	0.9863	0.0137	0.0761	0.9239	72.90	13.14	22.34
ICLN	0.9692	0.0308	0.0737	0.9263	32.45	13.57	21.99

Notes: This Table provides results of the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,t} + \beta_{i,t} R_{t-1} + \gamma_{i,t} \Delta VXXLE_t + \phi_{i,t} RPSE_t + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t and $RPSE_t$ is the return of the PSE index at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. In addition, RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

Table 10
Impact of the VXXLE controlling for oil price shocks.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	WTI	VXXLE	Sigma	χ^2 test
PBW	S1	−0.030 (0.33)	0.068*** (0.00)	0.091*** (0.00)	−0.470*** (0.00)	1.061*** (0.00)	132.70***
	S2	−0.056 (0.40)	0.029 (0.32)	0.014 (0.70)	−0.811*** (0.00)	1.632*** (0.00)	
PBD	S1	0.124*** (0.00)	−0.011 (0.80)	0.031 (0.11)	−0.200*** (0.00)	0.631*** (0.00)	122.53***
	S2	−0.083** (0.02)	−0.003 (0.86)	0.038** (0.02)	−0.679*** (0.00)	1.094*** (0.00)	
ICLN	S1	0.028 (0.46)	−0.069 (0.14)	0.049** (0.04)	−0.339*** (0.00)	0.886*** (0.00)	109.57***
	S2	−0.098* (0.06)	0.015 (0.51)	0.071*** (0.00)	−0.667*** (0.00)	1.336*** (0.00)	
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9951	0.0049	0.0095	0.9905	203.36	105.59	23.35
PBD	0.9431	0.0569	0.0295	0.9705	17.58	33.88	26.45
ICLN	0.9846	0.0154	0.0127	0.9873	65.04	78.46	24.79

Table 11
Impact of the VXXLE controlling for oil volatility (OVX) shocks.

Panel A: Estimated coefficients							
Index	State	Constant	AR (1)	OVX	VXXLE	Sigma	χ^2 test
PBW	S1	−0.023 (0.46)	0.074*** (0.00)	0.034 (0.46)	−0.638*** (0.00)	1.051** (0.03)	155.16***
	S2	−0.064 (0.35)	0.015 (0.59)	−0.083** (0.02)	−0.654*** (0.00)	1.665*** (0.00)	
PBD	S1	−0.260** (0.03)	−0.089* (0.07)	−0.043 (0.22)	−0.509*** (0.00)	1.733*** (0.00)	173.72***
	S2	0.024 (0.28)	0.035* (0.09)	−0.010 (0.62)	−0.583*** (0.00)	0.843*** (0.00)	
ICLN	S1	0.022 (0.53)	−0.075** (0.04)	0.014 (0.51)	−0.363*** (0.00)	0.886*** (0.00)	128.91***
	S2	−0.089** (0.04)	0.013 (0.56)	−0.034 (0.22)	−0.685*** (0.00)	1.336*** (0.00)	
Panel B: Transition probabilities and expected durations							
Index	P11	P12	P21	P22	DU1	DU2	RCM
PBW	0.9933	0.0067	0.0123	0.9876	150.23	81.25	20.74
PBD	0.9196	0.0804	0.0156	0.9844	12.44	63.95	22.67
ICLN	0.9878	0.0122	0.0101	0.9899	81.65	99.03	24.65

They confirm the significant negative effect of the VXXLE on clean energy ETFs. Regime 1 remains the high volatility state, except for the PBD index. However, unlike in the full sample analyses, the effect of the VXXLE is higher in regime 2 than regime 1 for PBW and ICLN. Therefore, for the low volatility regime, the aforesaid linkage gets stronger [31]. shows that during the sub-period analyses, the risk return relationship tends to tighten in a low-volatility trading environment. One possible reason could be that options traders respond aggressively to decreases in process during low-volatility periods by bidding up implied volatility, whereas options traders hesitate to do the same in continued high-volatility periods.

4.3.2. Robustness checking

To verify whether the findings presented in Table 4 are robust, we employ two alternative measures of volatility used in previous studies [22]. The first alternative measure is the realized volatility (RV) proxied by the squared returns of another energy sector ETF index, namely the XLE.⁴ The second alternative measure is the conditional volatility based on GARCH models. Given that squared returns appear to be a noisy measure of volatility, we estimate an AR (1)–GARCH(1,1) model for the XLE index. The results of the robustness analyses are shown in Tables 7 and 8. They confirm an asymmetric association between energy sector volatility shocks and clean energy ETFs.

Overall, our findings are robust, given that they remain extremely consistent irrespective of the sample size and measures used.

4.3.3. Further analyses with control variables

Previous studies provide empirical evidence that clean energy assets are sensitive to fluctuations in oil and technology stock prices [8,9,11,15,27,32,33]. For example, Sadorsky [8] argues that as investors view alternative energy companies as similar to other high technology companies, clean energy asset returns are substantially influenced by changes in technology stock prices. Recently, Song et al. [33] show that clean energy stocks are highly sensitive to oil price shocks. Another stand of literature shows that OVX exerts a significant impact on clean energy stock indices [12,1]. These studies provide evidence that crude oil volatility and clean energy stock price volatility are positively correlated.

In this section, we examine the impact of the VXXLE on clean energy assets after controlling for the effects of WTI oil prices, OVX, and technology stock prices proxied by the NYSE Arca Technology Index (PSE). Note that, since the WTI, OVX and PSE indices are highly correlated with the VXXLE, orthogonalized returns on these indices are considered. To this end, a two-step procedure is adopted. In the first step, the returns on WTI/OVX/PSE are regressed on the VXXLE to gauge its effect on these indices. In the second step, the residuals are estimated and incorporated into the MRS model along with the VXXLE in three separate analyses.

The results of these additional analyses are reported in Tables 9–11, where several interesting outcomes are observed. First, the impact of the VXXLE remains significant in each case considered. Second, the technology stock prices have a positive influence on the clean energy asset returns and, more importantly, the magnitude of this impact is higher than that of the VXXLE (see Table 9). Third, although the effect of WTI oil prices is significant in the majority of cases, the VXXLE impacts more than the oil prices as evidenced by the size of this effect (see Table 10). Fourth, the impact

⁴ XLE data are extracted from DataStream.

Table 12
Risk transmission relationship.

	PBW	PBD	ICLN	USO
Panel A: $RV_t = \delta_0 + \delta_1 RV_{t-1} + \varepsilon_t$				
Constant	2.15***	1.44***	1.81***	3.01***
RV_{t-1}	0.22***	0.21***	0.18***	0.20***
R^2	5.2%	4.6%	3.3%	4.1%
F-statistic	109.05***	94.71***	68.55***	84.53***
Panel B $RV_t = \delta_0 + \delta_1 RV_{t-1} + \delta_2 \Delta VXXLE_t + \varepsilon_t$:				
Constant	2.11***	1.42***	1.80***	2.98***
RV_{t-1}	0.24***	0.23***	0.19***	0.21***
$\Delta VXXLE_t$	0.90***	0.71***	0.73***	0.83***
R^2	10.7%	10.5%	8.6%	6.7%
F-statistic	117.75***	114.88***	92.04***	70.77***
Panel C $RV_t = \delta_0 + \delta_1 RV_{t-1} + \delta_2 \Delta OVX_t + \varepsilon_t$:				
Constant	2.15***	1.44***	1.82***	2.96***
RV_{t-1}	0.23***	0.22***	0.18***	0.22***
ΔOVX_t	0.54***	0.37***	0.39***	1.25***
R^2	8.5%	7.3%	5.8%	13.9%
F-statistic	91.20***	76.96***	60.89***	158.01***
Panel D $RV_t = \delta_0 + \delta_1 RV_{t-1} + \delta_2 RV_PSE_t + \varepsilon_t$:				
Constant	1.08***	0.56***	0.96***	2.51***
RV_{t-1}	0.14***	0.09***	0.10***	0.19***
RV_PSE_t	1.11***	0.93***	0.88***	0.45***
R^2	37.2%	41.8%	32.5%	7.1%
F-statistic	579.24***	701.72***	470.62***	75.25***

Notes: ***, ** and * indicate statistical significance at 1%, 5% and 10% levels respectively.

of the OVX appears to be insignificant in most cases, suggesting that the VXXLE captures variations in clean energy ETFs more than the crude oil price volatility (see Table 11). This last finding contradicts those reported by Dutta [26] and Ahmad et al. [3]; who show that clean energy indices react significantly to OVX shocks. The most likely explanation could be that crude oil and renewable energy no longer compete in the same markets. In fact, crude oil is mainly used to produce transportation fuels, while alternative energy is principally utilized to generate electricity. Therefore, by not being direct substitutes, when the price of one of them falls, the demand for the other does not necessarily decrease.⁵ Besides, we employ a non-linear framework, while the papers cited above adopt a linear model. Given that financial time series are usually non-linear in nature, adopting linear specifications could be misleading [34]. Although Dutta [26] conducts some non-linear or asymmetric analyses by examining whether positive or negative oil shocks matter for clean energy assets, his approach is essentially linear. Hence this research emphasizes the need to employ a non-linear framework to precisely detect the association among the variables under study.

Overall, our analysis concludes that the information in the VXXLE is more important than traditional oil prices or oil market volatility for modelling the returns of clean energy ETFs. In addition to the VXXLE, technology stock prices also play a major role in determining the movements of clean energy asset prices.

4.4. Risk transmission results

The estimates of models 5–8 are presented in Table 12. These findings show that the information content of VXXLE, OVX and RV_PSE play a pivotal role in explaining the variance of the ETFs under consideration. For example, in case of the PBW index, the R^2 statistics amount to 5.2%, 10.7%, 8.5% and 37.2% for the baseline, RV_VXXLE , RV_OVX and RV_PSE models, respectively. Hence, VXXLE, OVX and RV_PSE provide additional information beyond that

contained in the historical volatilities of these ETFs. We further observe that RV_PSE affects the volatility of clean energy ETFs more than VXXLE or OVX implying that clean energy assets are mainly sensitive to changes in RV_PSE . However, for the USO index, the impact of OVX is higher than the others. We thus report that energy sector ETFs and clean energy ETFs react differently to changes in OVX, VXXLE and RV_PSE . The possible explanation is that, while the price of oil is the primary driver of how the stock prices of fossil fuel companies are perceived, clean energy asset prices are more influenced by technology stock prices than crude oil volatility. This finding indicates that market participants view investment in clean energy stocks as more closely related to general movements in the technology sector rather than fluctuations in oil prices. Therefore, for clean energy assets, crude oil volatility shocks are not as important as shocks to technology stock price volatility. The findings in Table 12 show that the VXXLE has a better predictive power than the OVX. One would expect that, as fossil fuel and clean energy stocks are generally viewed as competing assets, stock prices of renewable energy firms would be affected by both positive and negative news stemming from the energy sector stock market [4]. Hence, the VXXLE, which captures 30-day market uncertainty for energy sector stocks, generates more accurate volatility estimates than the crude oil OVX. The energy sector ETF (i.e., USO), on the other hand, is more sensitive to OVX than VXXLE, which confirms that traditional energy assets are mainly influenced by variations in global oil prices. Overall, we show that traditional energy assets and clean energy assets have different reaction patterns to energy sector volatility.

5. Discussion

It is well documented that studying the risk-return link for the global clean energy equity markets is important, because proper knowledge about the linkage between an asset and its volatility (risk) is fundamental to understanding financial market uncertainty. Many studies use realized volatility (RV) to measure market risk, while another strand of literature shows that implied volatility indices such as the VIX perform better than other historical

⁵ Only liquid transport biofuels, e.g. biodiesel and bioethanol, directly compete with crude oil products.

measures (e.g., RV) at predicting stock market uncertainty [35].⁶ The literature on the risk-return nexus is growing, because holding VIX derivatives has been shown to help in hedging the risk associated with stock prices. Nevertheless, existing empirical work has not paid sufficient attention to this relationship for clean energy equities. The few exceptions include Ahmad et al. [3] and Dutta et al. [36] who document that commodity volatility indices such as oil VIX, gold VIX and silver VIX can diversify the risk of clean energy assets.

Throughout this paper, we extend this scarce literature by using the information content of the VXXLE for predicting the prices of clean energy assets. Our results indicate that the relationship between the VXXLE and clean energy ETFs is asymmetric, showing that negative returns lead to higher volatility than positive returns of the same size. This finding is comparable to previous studies on US equities. For example, Mencía and Sentana [37] claim that investors trade options on underlying assets in order to buy protection during crisis periods, which increases the value of VIX indices.

Our empirical investigation shows that clean energy assets are sensitive to changes in technology stock prices, while oil price volatility does not much matter for these ETFs. This finding shows that oil price is not the main driver of clean energy assets, as market participants might view these firms as similar to other high technology companies. On the whole, technology stock prices along with the VXXLE play a crucial role in determining the movements of clean energy asset prices. Therefore, investors holding assets in renewable energy markets should closely observe fluctuations in technology stock prices and energy sector volatility indices. Furthermore, the risk transmission analysis shows that the information content of VXXLE, OVX and RV_PSE play a pivotal role in explaining the variance of clean energy ETFs.

6. Conclusions

In this study, we examine the impact of energy sector volatility on clean energy ETFs. Unlike previous studies which mainly employ linear models to study the dynamics of the association between clean energy and other asset classes, we apply the Markov regime switching regression. Notably, we use the energy sector implied volatility index (VXXLE) to measure the risk or uncertainty of energy sector firms. Our findings reveal that the VXXLE has a negative impact on various clean energy ETFs, suggesting that when volatility levels are high for energy sector companies, a drop is likely to be observed in clean energy ETFs. Further analyses indicate an asymmetric association. We also show that for the high volatility regime, the association between energy sector volatilities and renewable energy ETFs gets stronger than for the low volatility regime. Our above-mentioned results remain unchanged controlling for the effects of oil prices, crude oil volatility, and technology stock prices. In addition, the results demonstrate that changes in the levels of the VXXLE substantially impact the realized volatility of these ETFs.

Our analyses offer significant implications for investors and policymakers. The findings of this study could receive special attention from current and future investors in clean energy sectors. Given that rising worries about energy security and climate change shift market participants towards ethical investments, our results

might help investors who aim to decarbonize their portfolios by including more clean energy equities. In fact, our findings could help investors understand the risk associated with clean energy assets and hedge portfolio risk. Given that energy sector volatility and alternative energy firms' asset returns are inversely associated, investors and risk managers can consider potential diversification of the downside risk of clean energy markets using the VXXLE. The results concerning the risk transmission between the VXXLE and the volatility of clean energy ETFs, matter for option pricing and risk spillover inferences, indicating the need to follow the path of the VXXLE when predicting the volatility of clean energy ETFs. For policymakers, our main findings are important in the sense that an upward trend in energy sector volatility does not provide incentive to invest in the clean energy industry, while a downward trend could increase clean energy investment without the need for specific support from energy policies. The results are also relevant to the formulation of policies seeking to raise the attractiveness and stability of investments in clean energy assets, which are important for the sustainable development agenda.

Our analyses offer scope for future research. For example, future studies could examine whether energy market uncertainty impacts clean energy sub-sectors. In this regard, time-varying correlations among implied volatilities and various sub-sectors of the clean energy equity markets could be explored to make inferences about portfolio implications and risk management at the disaggregated level.

Credit author statement

Anupam Dutta: Conceptualization; Methodology; Formal analysis; Writing - original draft. Elie Bouri: Project administration; Validation; Writing - original draft; Final Revision, Tareq Saeed: Writing - Review, Editing, Final Revision. Xuan Vinh Vo: Writing - Review, Final Revision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2020.118657>.

Notes: This table shows the results of the RALS-LM unit root test. $\hat{k}$ indicates the optimal number of lags selected using a general to specific procedure, while $\hat{T}_B$ denotes the location of structural breaks. The corresponding critical values for $\rho^2 = 0.1, \dots, 0.9$ can be found in Refs. [38]. The test-statistics for RALS-LM unit root tests appear to be invariant to the location of breaks. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

Notes The results are based on the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,t} + \beta_{i,t}R_{t-1} + \gamma_{1i,t}\Delta VXXLE_t^+ + \gamma_{2i,t}\Delta VXXLE_t^- + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t and $\Delta VXXLE_t^+ = \max(\Delta VXXLE_t, 0)$, $\Delta VXXLE_t^- = \min(\Delta VXXLE_t, 0)$. Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as D_{ui} , i.e., D_{U1} for state 1 and D_{U2} for state 2. The χ^2 test is performed to examine whether the state variances are

⁶ One possible explanation is that the VIX reflects the forward-looking measure of uncertainty. [27]; for example, argue that the VIX is of paramount importance to investors on a daily basis for decisions related to trading. Given that the VIX contains market expectations, this measure appears to be a useful instrument for predicting market risk. Overall, the information in the VIX reflects the risk and return patterns.

equal. The asymmetric impact ($H_0: \gamma_1 = \gamma_2$) is investigated by applying the student's t -test. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

Notes: The subsample analyses are based on the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t}R_{t-1} + \gamma_{i,r_t}\Delta VXXLE_t + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

Notes: This Table provides results of the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t}R_{t-1} + \gamma_{i,r_t}RV_t + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t and RV_t is the realized volatility of the XLE index at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

Notes: This Table provides results of the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t}R_{t-1} + \gamma_{i,r_t}\Delta VXXLE_t + \phi_{i,r_t}RWIT_t + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t and $RWIT_t$ is the return of the WTI oil index at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

Notes: This Table provides results of the Markov regime switching (MRS) model of the form: $R_{i,t} = \alpha_{i,r_t} + \beta_{i,r_t}R_{t-1} + \gamma_{i,r_t}\Delta VXXLE_t + \phi_{i,r_t}\Delta OVX_t + u_{i,t}$, where $R_{i,t}$ refers to the return for clean energy assets at time t and ΔOVX_t is the first-order difference of OVX at time t . Panel A shows the estimates of the MRS process and Panel B shows the transition probabilities and expected durations. RCM is the regime classification measure. The transition probabilities are reported as P_{ij} . The expected duration of being in state i is reported as DU_i , i.e., DU_1 for state 1 and DU_2 for state 2. The χ^2 test is performed to examine whether the state variances are equal. p -values are given in parentheses. ***, ** and * indicate statistically significant results at 1%, 5% and 10% levels respectively.

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